What is Linear Regression?

Definition

Linear regression assumes the output is a linear combination of the input features.

Mathematical Form:

$$y = w_0 + w_1 x_1 + w_2 x_2 + \cdots + w_d x_d \quad (1)$$

where:

- $\mathbf{x} = [x_1, x_2, \ldots, x_d]^T$ are the input features
- $\mathbf{w} = [w_1, w_2, \ldots, w_d]^T$ are the weights
- w_0 is the bias term

Examples of Linear Regression

1. Salary Prediction
   - Salary $= w_0 + w_1 \times$ Years of Experience
2. Crop Yield Prediction
   - Yield $= w_0 + w_1 \times$ Rainfall $+ w_2 \times$ Fertilizer
3. Energy Consumption
   - Energy $= w_0 + w_1 \times$ Temperature $+ w_2 \times$ Hour $+ w_3 \times$ Day

The relationship between inputs and output is assumed to be linear.

Table of Contents

The Four Components

A complete machine learning system consists of:

1. Data: d-dimensional input vectors and target outputs
2. Model: Single neuron implementing linear function
3. Objective Function: Measures prediction error
4. Learning Algorithm: Gradient descent to optimize weights

Matrix Form of Data

For efficient computation, we organize data into matrices:

Input Data = Augmented Features:

Add bias term: $\tilde{\mathbf{x}}^{(i)} = [1, x_1^{(i)}, \ldots, x_d^{(i)}]^T$

Each row = one training example

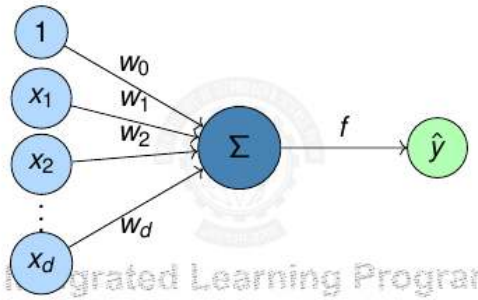
Design Matrix: $\mathbf{X} \in \mathbb{R}^{N \times (d+1)}$ $\mathbf{X} = \begin{bmatrix} 1 & x_1^{(1)} & \cdots & x_d^{(1)} \\ 1 & x_1^{(2)} & \cdots & x_d^{(2)} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_1^{(N)} & \cdots & x_d^{(N)} \end{bmatrix}$

Target Vector: $\mathbf{y} \in \mathbb{R}^N$ $\mathbf{y} = \begin{bmatrix} y^{(1)} & y^{(2)} & \ldots & y^{(N)} \end{bmatrix}^\top$

Weight Vector: $\mathbf{w} \in \mathbb{R}^{d+1}$ $\mathbf{w} = \begin{bmatrix} w_0 & w_1 & w_2 & \ldots & w_d \end{bmatrix}^\top$

Linear Model as a Single Neuron

Model linear regression as a single artificial neuron:



- Input: d-dimensional feature vector $\mathbf{x} \in \mathbb{R}^d$
- Output: Single predicted value $\hat{y} \in \mathbb{R}$

Linear Model Prediction

Model: A single neuron computes a weighted sum and applied identity activation function.

Prediction for input $\mathbf{x}$:

$$\hat{y} = f(\mathbf{w}^T\mathbf{x}) = f(w_0 + w_1x_1 + w_2x_2 + \cdots + w_dx_d) \tag{2}$$

where:

- $\mathbf{w} = [w_0, w_1, \ldots, w_d]^T \in \mathbb{R}^{d+1}$ are the model parameters
- $\mathbf{x} = [1, x_1, \ldots, x_d]^T \in \mathbb{R}^{d+1}$ is the input
- $f(\cdot)$ is the identity activation $f(z) = z$

Vectorized form: for all predictions

$$\hat{\mathbf{y}} = \mathbf{X}\mathbf{w} \tag{3}$$

Identity Activation Function

Identity Activation Function

$$f(z) = z \tag{4}$$

Properties of Identity Function

- Output: Continuous, $(-\infty, \infty)$
- Differentiable everywhere: $f'(z) = 1$
- Allows output to take any real value
- Gradient flows unchanged through the neuron
- Perfect for predicting continuous targets

Why identity for regression?

- Target values are continuous (prices, temperatures, etc.)
- Need to predict any real number, not just binary { 0,1 }
- Differentiability enables gradient-based optimization
- No information loss from input to output

Objective Function: Squared Error Loss

We need to measure how well our model fits the data.

Loss for single example:

$$\ell\left(\mathbf{w}; \mathbf{x}^{(i)}, y^{(i)}\right) = \frac{1}{2}\left(\hat{y}^{(i)} - y^{(i)}\right)^2 = \frac{1}{2}\left(\mathbf{w}^T\mathbf{x}^{(i)} - y^{(i)}\right)^2 \tag{5}$$

Total Loss (Mean Squared Error):

$$J(\mathbf{w}) = \frac{1}{N}\sum_{i=1}^{N}\frac{1}{2}\left(\mathbf{w}^T\mathbf{x}^{(i)} - y^{(i)}\right)^2 \tag{6}$$

Goal: Find $\mathbf{w}^*$ that minimizes $J(\mathbf{w})$

$$\mathbf{w}^* = \arg\min_{\mathbf{w}} J(\mathbf{w}) \tag{7}$$

Why Squared Error?

Advantages of squared error loss:

- Differentiable: Smooth everywhere, easy to optimize
- Convex: Single global minimum (for linear models)
- Penalizes large errors: Quadratic growth
- Statistical interpretation: Maximum likelihood under Gaussian noise

Vector Form:

$$J(\mathbf{w}) = \frac{1}{2N}\|\mathbf{X}\mathbf{w} - \mathbf{y}\|^2 = \frac{1}{2N}(\mathbf{X}\mathbf{w} - \mathbf{y})^T(\mathbf{X}\mathbf{w} - \mathbf{y}) \tag{8}$$

where $\mathbf{X} \in \mathbb{R}^{N\times(d+1)}$ is the design matrix and $\mathbf{y} \in \mathbb{R}^N$ is the target vector.

Batch Gradient Descent Algorithm

Algorithm 1: Gradient Descent for Linear Regression

Input: Dataset $\mathcal{D} = \{(\mathbf{x}^{(i)}, y^{(i)})\}_{i=1}^N$, learning rate η, tolerance ϵ
Output: Learned weights $\mathbf{w}$

Initialize $\mathbf{w}^{(0)} = \mathbf{0}$ (or random);
$t \leftarrow 0$;
while *not converged* **do**
  Compute gradient: $\nabla J(\mathbf{w}^{(t)}) = \frac{1}{N}\mathbf{X}^T(\mathbf{X}\mathbf{w}^{(t)} - \mathbf{y})$;
  Update weights: $\mathbf{w}^{(t+1)} = \mathbf{w}^{(t)} - \eta\nabla J(\mathbf{w}^{(t)})$;
  if $\|\nabla J(\mathbf{w}^{(t+1)})\| < \epsilon$ **then**
    break // Converged
  $t \leftarrow t + 1$;
return $\mathbf{w}^{(t+1)}$;

Computing the Gradient

Gradient of squared error loss:

$$\nabla J(\mathbf{w}) = \frac{\partial J}{\partial \mathbf{w}} = \frac{1}{N}\sum_{i=1}^{N}\left(\mathbf{w}^T\mathbf{x}^{(i)} - y^{(i)}\right)\mathbf{x}^{(i)} \tag{9}$$

Matrix Form:

$$\nabla J(\mathbf{w}) = \frac{1}{N}\mathbf{X}^T(\mathbf{X}\mathbf{w} - \mathbf{y}) \tag{10}$$

where $\mathbf{X}^T \in \mathbb{R}^{(d+1)\times N}$, $\mathbf{X}\mathbf{w} - \mathbf{y} \in \mathbb{R}^N$, so $\nabla J \in \mathbb{R}^{d+1}$

Gradient Descent Update Rule

Update equation:

$$\mathbf{w}^{(t+1)} = \mathbf{w}^{(t)} - \frac{\eta}{N}\mathbf{X}^T\left(\mathbf{X}\mathbf{w}^{(t)} - \mathbf{y}\right) \tag{11}$$

Key Parameters:

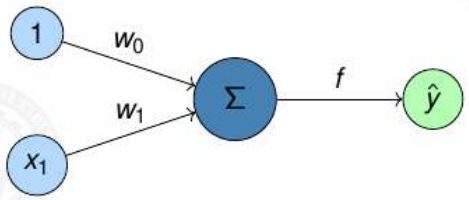
- Learning rate η: Controls step size
  - Too large: May overshoot minimum
  - Too small: Slow convergence
- Stopping criterion: $\|\nabla J(\mathbf{w})\| < \epsilon$ or max iterations

Table of Contents

Numerical Example: Setup

Tiny Dataset: Predict house price from size

Size (x_1)	Price (y)
1	2
2	4
3	5



Matrix Formulation:

$$\mathbf{X} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \in \mathbb{R}^{3\times 2}, \quad \mathbf{y} = \begin{bmatrix} 2 \\ 4 \\ 5 \end{bmatrix} \in \mathbb{R}^3, \quad \mathbf{w} = \begin{bmatrix} w_0 \\ w_1 \end{bmatrix} \in \mathbb{R}^2$$

Numerical Example: Initialization

Hyperparameters: $\eta = 0.1 \quad N = 3$

Initialize weights: $\mathbf{w}^{(0)} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

Vectorized Prediction: $\hat{\mathbf{y}} = \mathbf{X}\mathbf{w}$

Initial predictions: $\hat{\mathbf{y}}^{(0)} = \mathbf{X}\mathbf{w}^{(0)} = \begin{bmatrix} 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

Loss Function: $J(\mathbf{w}) = \frac{1}{2N}\|\mathbf{X}\mathbf{w} - \mathbf{y}\|^2$

Initial loss: $J(\mathbf{w}^{(0)}) = \frac{1}{6}\left\| \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} - \begin{bmatrix} 2 \\ 4 \\ 5 \end{bmatrix} \right\|^2 = \frac{1}{6}(4 + 16 + 25) = 7.5$

Numerical Example: Computing Gradient

Error vector: $\mathbf{e}^{(0)} = \hat{\mathbf{y}}^{(0)} - \mathbf{y} = \begin{bmatrix}0\\0\\0\end{bmatrix} - \begin{bmatrix}2\\4\\5\end{bmatrix} = \begin{bmatrix}-2\\-4\\-5\end{bmatrix}$

Gradient computation: $\nabla J(\mathbf{w}^{(0)}) = \frac{1}{N}\mathbf{X}^T\mathbf{e}^{(0)}$

$$\nabla J(\mathbf{w}^{(0)}) = \frac{1}{3}\begin{bmatrix}1 & 1 & 1\\1 & 2 & 3\end{bmatrix}\begin{bmatrix}-2\\-4\\-5\end{bmatrix}$$

$$= \frac{1}{3}\begin{bmatrix}-2-4-5\\-2-8-15\end{bmatrix} = \begin{bmatrix}-3.67\\-8.33\end{bmatrix}$$

Numerical Example: Weight Update

Gradient descent update: $\mathbf{w}^{(1)} = \mathbf{w}^{(0)} - \eta\nabla J(\mathbf{w}^{(0)})$

Computing new weights: $\mathbf{w}^{(1)} = \begin{bmatrix}0\\0\end{bmatrix} - 0.1\begin{bmatrix}-3.67\\-8.33\end{bmatrix} = \begin{bmatrix}0.367\\0.833\end{bmatrix}$

New predictions: $\hat{\mathbf{y}}^{(1)} = \mathbf{X}\mathbf{w}^{(1)}$

$$\hat{\mathbf{y}}^{(1)} = \begin{bmatrix}0.367 & 0.833\end{bmatrix}\begin{bmatrix}1 & 1 & 1\\1 & 2 & 3\end{bmatrix} = \begin{bmatrix}1.20\\2.03\\2.87\end{bmatrix}$$

Numerical Example: New Loss

New error vector: $\mathbf{e}^{(1)} = \hat{\mathbf{y}}^{(1)} - \mathbf{y} = \begin{bmatrix}1.20\\2.03\\2.87\end{bmatrix} - \begin{bmatrix}2\\4\\5\end{bmatrix} = \begin{bmatrix}-0.80\\-1.97\\-2.13\end{bmatrix}$

Loss after one iteration: $J(\mathbf{w}^{(1)}) = \frac{1}{6}\|\mathbf{e}^{(1)}\|^2 = \frac{1}{6}(0.64 + 3.88 + 4.54) \approx 1.51$

Progress:

- Initial loss: $J(\mathbf{w}^{(0)}) = 7.5$
- After 1 iteration: $J(\mathbf{w}^{(1)}) = 1.51$
- **80% reduction!** ✓

Continue until $\|\nabla J(\mathbf{w})\| < \epsilon$ or max iterations reached

Computational Graph for One Gradient Descent Step

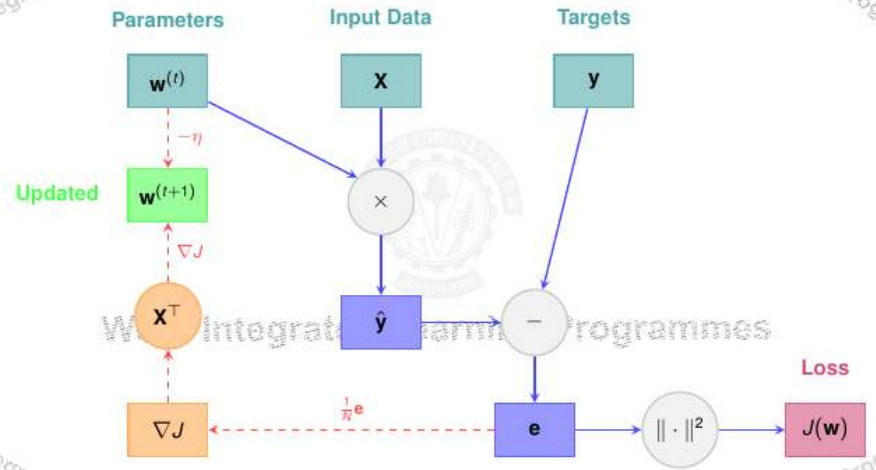


Table of Contents

Model Evaluation: Training vs Test Error

Key Principle: Evaluate on unseen data

Data Split:

- Training set: Used to learn weights $\mathbf{w}$ (typically 90-99%)
- Test set: Used to evaluate generalization (typically 1-10%)

$$\text{Training Error:} \quad J_{\text{train}} = \frac{1}{N_{\text{train}}} \sum_{i \in \text{train}} \left(\hat{y}^{(i)} - y^{(i)} \right)^2$$

$$\text{Test Error:} \quad J_{\text{test}} = \frac{1}{N_{\text{test}}} \sum_{i \in \text{test}} \left(\hat{y}^{(i)} - y^{(i)} \right)^2$$

Goal: Minimize test error (generalization performance)

Evaluation Metrics

1. Mean Squared Error (MSE):

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^{N} \left(\hat{y}^{(i)} - y^{(i)} \right)^2 \tag{12}$$

2. Root Mean Squared Error (RMSE):

$$\text{RMSE} = \sqrt{\text{MSE}} = \sqrt{\frac{1}{N} \sum_{i=1}^{N} \left(\hat{y}^{(i)} - y^{(i)} \right)^2} \tag{13}$$

3. Mean Absolute Error (MAE):(Less sensitive to outliers than MSE)

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^{N} | \, y^{(i)} - \hat{y}^{(i)} \, | \tag{14}$$

R^2 Score (Coefficient of Determination)

Definition: Proportion of variance explained by the model

$$R^2 = 1 - \frac{\sum_{i=1}^{N} (y^{(i)} - \hat{y}^{(i)})^2}{\sum_{i=1}^{N} (y^{(i)} - \bar{y})^2} = 1 - \frac{\text{SS}_{\text{residual}}}{\text{SS}_{\text{total}}} \tag{15}$$

where $\bar{y} = \frac{1}{N} \sum_{i=1}^{N} y^{(i)}$ is the mean.

Interpretation:

- $R^2 = 1$: Perfect fit (all variance explained)
- $R^2 = 0$: Model no better than predicting mean
- $R^2 < 0$: Model worse than baseline (possible on test set)
- Typical: $0.7 < R^2 < 0.9$ indicates good fit